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En un espacio de Hilbert, convergencia débil y en norma implica convergencia fuerte

Proposición 1. *Sea H un espacio de Hilbert real y $(x_n)_{n\in\mathbb{N}} \subset H$, entonces*

$$x_n \rightharpoonup x \wedge \|x_n\| \to \|x\| \implies x_n \to x \text{ en } H.$$

***Demostración*:** Basta ver que $\|x_n - x\| \to 0$:

$$\begin{aligned}\|x_n - x\|^2 &= \langle x_n - x, x_n - x\rangle = \|x_n\|^2 - 2\langle x_n, x\rangle + \|x\|^2 \\ &= \|x_n\|^2 - 2\left(L_x(x_n)\right) + \|x\|^2 .\end{aligned}$$

Al tomar límites cuando $n \to \infty$ en la expresión anterior, obtenemos que

$$\|x_n - x\|^2 \xrightarrow{n\to\infty} \|x\|^2 - 2\left(L_x(x)\right) + \|x\|^2 = \|x\|^2 - 2\|x\|^2 + \|x\|^2 = 0.$$

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